

AUTOMATED PRODUCE SCANNER

Hardware Setup and Assembly

Real-Time Pricing and Packaging System

Portfolio / Technical Documentation Edition

Acknowledgement

The successful completion of this project would not have been possible without the collective effort, knowledge, cooperation, and support of the individuals who contributed throughout its development.

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May God bless us all.

1. Hardware Setup Overview

The Automated Produce Scanner integrates sensing, processing, user-interface, transaction-output, electromechanical, mechanical, and power components into a single embedded prototype. The Raspberry Pi 4 Model B functions as the central processing unit and coordinates the camera, weighing system, LCD, thermal printer, and automated sealing mechanism.

The hardware selection considered component compatibility, weight capacity, lighting conditions, packaging material, sensor accuracy, cost, and availability. The setup described here is based on the submitted capstone thesis.

2. Hardware Architecture

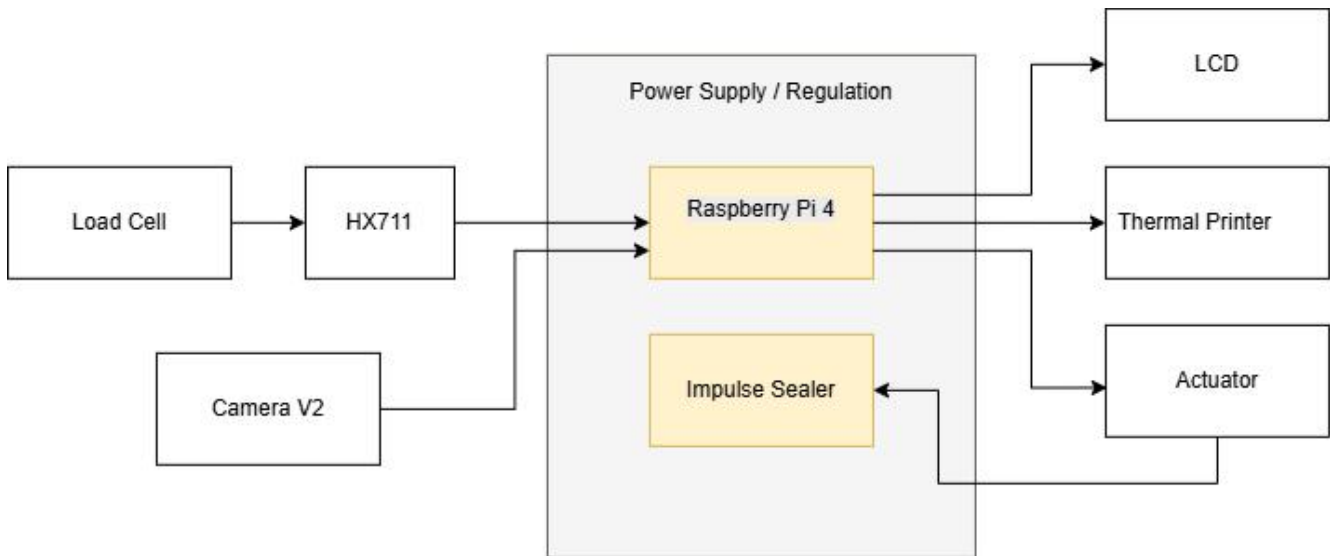


Figure 1. Functional hardware connection arrangement.

This is a functional connection map rather than a pin-level electrical schematic. The thesis describes the functional relationships among the components but does not provide a complete authoritative GPIO pinout.

3. Hardware Components and Functions

Component	Function
Raspberry Pi 4 Model B (4 GB)	Central processing unit; runs image detection, processes weight data, calculates prices, and coordinates outputs.
Raspberry Pi Camera Module V2 (IMX219)	Captures produce images for machine-learning-based identification.
Load Cell	Measures the weight of produce placed on the weighing platform.
HX711 Module	Amplifies and converts the load-cell signal for

	Raspberry Pi processing.
LCD with I ² C Module	Displays vegetable type, detection accuracy, weight, and price.
Thermal Printer	Prints barcode receipts or price labels containing transaction information.
Impulse Sealer	Seals produce in heat-sealable LDPE plastic bags.
Linear Actuator	Pushes and pulls the impulse sealer during automated packaging.
Push Buttons	Provide user controls including tare/reset and transaction confirmation.
Power Supply	Provides electrical power to the Raspberry Pi and other hardware.
Buck Converter	Steps down higher DC voltage to a lower stable level for supported components.
Terminal Block	Organizes and secures wiring connections.
Raspberry Pi Flex Cable	Connects the Camera Module V2 to the Raspberry Pi CSI interface.
Barcode Scanner	Reads the printed barcode to retrieve stored transaction data.
Mechanical Structure	Acrylic casing, 10 mm plywood, stainless angle bar, screws, and pulleys support the physical prototype.

4. Physical Hardware Arrangement

The physical arrangement places the camera above the produce platform to provide a consistent image capture area. The load cell is positioned beneath the weighing platform. The LCD is placed where the user can view the transaction information, while the thermal printer is positioned for label collection. The linear actuator and impulse sealer form the automated packaging section.

- Mount the camera at a fixed height above the produce platform.
- Secure the load cell beneath the weighing platform so the produce load is transferred consistently.
- Mount the LCD in a visible user-facing position.
- Install the thermal printer where printed labels can be accessed.
- Install the actuator and impulse sealer so the actuator can mechanically operate the sealer.
- Secure the Raspberry Pi and organize wiring using the terminal block.
- Keep wires away from moving mechanical parts.

5. Electrical and Functional Connections

Subsystem	Functional Connection
Camera	Camera Module V2 → Raspberry Pi CSI interface → image data.
Weighing	Load Cell → HX711 → Raspberry Pi → weight data.
Display	Raspberry Pi → LCD/I ² C module → user display.
Transaction Output	Raspberry Pi → Thermal Printer → barcode/label.
Packaging Control	Raspberry Pi → actuator control → Linear Actuator → Impulse Sealer.
User Controls	Push buttons → Raspberry Pi → tare/reset and transaction confirmation.
Power	Power supply → regulated/distributed power → system components; buck converter is used where voltage reduction is required.

Exact GPIO assignments are intentionally omitted. Before reconstructing the device, verify the final working prototype, source code, and any original wiring schematic.

6. Recommended Hardware Assembly Sequence

Prepare the enclosure and platform: Prepare the acrylic/plywood structure and establish locations for the camera, weighing platform, LCD, printer, and sealing mechanism.

Install the weighing platform: Mount the load cell beneath the platform and ensure mechanical loads are transferred correctly.

Install the camera: Mount the Raspberry Pi Camera Module V2 above the platform at the intended fixed position.

Install the Raspberry Pi: Secure the Raspberry Pi inside the enclosure and route connections cleanly.

Connect the weighing subsystem: Connect the load cell to the HX711 and the HX711 to the Raspberry Pi using the verified final wiring.

Install the LCD and buttons: Mount the LCD and install the documented user controls.

Install the printer: Mount and connect the thermal printer.

Build the sealing mechanism: Install the linear actuator and impulse sealer and verify mechanical travel.

Install power distribution: Connect the power supply and supporting regulation/distribution components.

Secure and inspect wiring: Check connections, strain relief, insulation, and clearance from moving parts.

Test individual modules: Test each subsystem independently before integration.

Integrate the complete device: Run the complete detection → weighing → pricing → printing → sealing sequence.

7. Individual Hardware Testing

Module	Test
Camera	Stable image capture and correct viewing area.
Load Cell + HX711	Calibration using known/precision weights and repeatable readings.
LCD	Correct display of produce type, accuracy, weight, and price.
Thermal Printer	Clear barcode/label printing and successful scanning.
Push Buttons	Correct tare/reset and confirmation behavior.
Linear Actuator	Correct travel and controlled movement.
Impulse Sealer	Consistent sealing of supported LDPE bags.
Power System	Stable operation and secure electrical connections.

8. Integrated Hardware Operation

During a normal transaction, a single produce item is placed on the weighing platform. The camera captures the item while the load cell measures its weight. The Raspberry Pi processes the image and obtains the produce classification. The weight and the corresponding database price are used to calculate the transaction amount. The LCD presents the result. After confirmation, the actuator operates the impulse sealer and the thermal printer generates the barcode label.

If no produce is detected, the system enters a waiting state. If multiple produce items are detected, the system enters an assorted-produce error state and waits until a valid single-produce condition is restored. The tare control resets the weighing system before measurement.

9. Bill of Materials

Component	Quantity	Listed Cost
Raspberry Pi 4 Model B	1	₱6,175.00
Load Cell with HX711 Module	1	₱435.00
Raspberry Pi Camera Module V2 (IMX219)	1	₱1,935.00
LCD with I ² C Module	1	₱70.00
Push Button	2	₱13.00 each
Resistor (330 Ω)	1	₱2.00

Linear Actuator	1	₱1,680.00
Heat Impulse Sealer	1	₱580.00
Power Supply	1	₱415.00
Barrel Jack	1	₱130.00
Stepper Motor Drive	1	₱340.00
Buck Converter	1	₱85.00
Terminal Block Set	20	₱88.00 listed total
RPi Flex Cable (1000 mm)	1	₱130.00
Thermal Printer	1	₱1,595.00
Thermal Sticker	1	₱210.00
Barcode Scanner	1	₱870.00
Stainless Angle Bar	15	₱205.00 listed total
Flat Head Screw Set	48	₱205.00 listed total
Acrylic Casing	1	₱3,600.00
Plywood (10 mm)	1	₱600.00
Pulley	2	₱35.00 listed total

The thesis reports a total materials cost of ₱19,411.00 for the prototype.

The thesis also lists a stepper motor drive but explicitly states that it was not part of the main implemented setup and may be used for future mechanical control.

10. Hardware Limitations

- The weighing system is documented as accommodating approximately 100 g to 3 kg.
- The system is designed to process one kind of vegetable at a time.
- The packaging mechanism is limited to supported heat-sealable LDPE plastic bags.
- The prototype operates sequentially rather than supporting parallel transactions.
- Weight readings may exhibit minor fluctuations caused by sensor sensitivity and load distribution.
- The sealing mechanism required calibration and hardware adjustment to improve consistency.

11. Safety and Maintenance

- Disconnect power before modifying or troubleshooting wiring.
- Keep hands and objects away from the actuator and heat-sealing area during operation.
- Allow the impulse sealer to cool before maintenance.
- Inspect terminal connections and wiring regularly.
- Keep the weighing platform free from unintended mechanical loads.
- Keep the camera lens and imaging area clean.

- Check actuator alignment and sealing position before continuous operation.

12. Documentation Note

This document is a portfolio-oriented technical adaptation of the submitted capstone thesis. It uses original functional diagrams/text structure rather than reproducing the thesis figures. The source thesis is identified as an unpublished capstone project. Exact GPIO pin assignments, wire colors, electrical ratings, and mechanical dimensions should only be documented after verification against the final physical prototype, source code, or original schematic.